

DANIEL XIANG

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ACADEMIC APPOINTMENT

Assistant Instructional Professor, *University of Chicago*

September 2023 – Present

EDUCATION

PhD in Statistics, *University of Chicago*

September 2017 – August 2023

Advisors: Peter McCullagh and Chao Gao

ScB in Applied Mathematics, *Brown University*

September 2013 – May 2017

Academic advisor: Bjorn Sandstede

RESEARCH INTERESTS

Empirical Bayes, Multiple hypothesis testing, Selective inference

RESEARCH

Publications

Xiang, D., Ignatiadis, N., & McCullagh, P. (2025). Interpretation of local false discovery rates under the zero assumption. *Statistical Science*, to appear. [arXiv](#).

Xiang, D., Soloff, J. A., & Fithian, W. (2026). A frequentist local false discovery rate. *Biometrika*. [arXiv](#).

Xiang, D., & Gao, C. (2026). Sharp phase transitions in high-dimensional changepoint detection. *Bernoulli*. [arXiv](#).

Tresoldi, M., Xiang, D., & McCullagh, P. (2024). Sparse-limit approximation for t -statistics. *Electronic Journal of Statistics*. [Journal](#). [arXiv](#).

Soloff, J. A., Xiang, D., & Fithian, W. (2024). The edge of discovery: Controlling the local false discovery rate at the margin. *Annals of Statistics*. [Journal](#). [arXiv](#).

Preprints

Xiang, D., Bogdan, M., Wallin, J., & Yekutieli, D. (2026). Hierarchical Bayesian estimation of covariance matrices. [arXiv](#).

Xiang, D., Fithian, W., Ignatiadis, N., Soloff, J., & Weinstein, A. (2026). Estimating the local false discovery rate under an unknown symmetric null. [arXiv](#).

Mostow, S., & Xiang, D. (2026). Adaptive procedures for boundary FDR control. [arXiv](#).

Gao, Z., Roquain, E., & Xiang, D. (2026). Reliable conformal novelty detection at the decision boundary. [arXiv](#).

Battey, H., McCullagh, P., & Xiang, D. (2025). Non-standard boundary behaviour in binary mixture models. [arXiv](#).

Xiang, D., & McCullagh, P. (2020). Permanental graphs. [arXiv](#).

TEACHING

UChicago

2017 – Present

Elementary Statistics (**Instructor**, Autumn 2026*).

Statistical Methods and Applications (**Instructor**, Spring 2020, Spring 2024, Winter 2025, Spring 2025, Winter 2026, Spring 2026).

Applied Regression Analysis (**Instructor**, Autumn 2023, Winter 2024, Autumn 2025, Autumn 2026*).

Hypothesis testing with Empirical Bayes (**Instructor**, Winter 2024, Winter 2025, Autumn 2025, Autumn 2026*).

Statistical Models and Methods (**Instructor**, Spring 2022, Autumn 2022, Autumn 2024).

PhD theory qualifying exam preparation (**Coach**, Summer 2022).

Statistical Theory and Methods (**TA**, Winter 2021), taught by Professor Rina Barber.

Bayesian Data Analysis (**TA**, Spring 2021), taught by Professor Fei Liu.

*Currently teaching

AWARDS

David Cox Research Prize (Royal Statistical Society, Autumn 2026).

First place team, Citadel Datathon (UC Berkeley, Autumn 2017).

Magna cum laude (Brown University, Spring 2017).

Rohn Truell Prize in Applied Math (Brown University, Spring 2017).

Phi Beta Kappa (Brown University, Spring 2016).

TALKS

Interpretation of local false discovery rates under the zero assumption. *The 13th International Conference on Multiple Comparison Procedures*, August 2025.

A frequentist local false discovery rate. *International Seminar on Selective Inference*, February 2025.

FDR at the boundary. *IMS International Conference on Statistics and Data Science (Nice, France)*, December 2024.

Frequentist local false discovery rates. *Tel Aviv University Statistics Seminar*, April 2023.

A frequentist perspective on the local false discovery rate. *The 12th International Conference on Multiple Comparison Procedures*, August 2022.